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Towards Optimal Guidance of Autonomous Swarm Drones in Dynamic Constrained Environments

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ABSTRACT

As autonomous drone swarms become increasingly important for complex missions, there remains a critical need for integrated approaches that can simultaneously handle task allocation and safe navigation in dynamic environments. This paper addresses the challenge of optimally allocating tasks and generating collision-free trajectories for drone swarms operating in obstacle-rich settings. Our proposed Swarm Allocation and Route Generation (SARG) framework integrates optimal task assignment with dynamically feasible trajectory planning, enabling efficient mission completion while ensuring safe navigation through complex 3D workspaces. Using quadrotors as our experimental platform, the framework incorporates both Drone-to-Obstacle and Drone-to-Drone collision avoidance algorithms, alongside a modified path planning algorithm that enhances simultaneous graph search efficiency. Our extensive experiments demonstrate that the SARG framework significantly improves performance over existing approaches. The SARG framework, while maintaining a 100% collision avoidance rate in dense environments, achieves a 21.6% reduction in the computation time of the simultaneous graph searching phase compared to conventional methods, contributing to overall system efficiency. These results establish SARG as a viable solution for real-world autonomous drone swarm applications in complex, dynamic settings. Supporting Information, including animated simulations, are available at <https://youtu.be/56oabPTUz4g>.

1 | Introduction

Autonomous Drones, commonly known as Unmanned Aerial Vehicles (UAVs), have evolved significantly over recent decades and have found widespread use in both military and civilian applications (Chai et al. 2024). Their high flexibility, versatility, agility, low risk, low cost, and ease of deployment make them ideal for tasks such as search and rescue, remote sensing, surveillance, 3D mapping, object grasping, aerial manipulation, underground mine exploration, and infrastructure inspection (Bale et al. 2024; Zrelli et al. 2024; Alqudsi et al. 2021a). These versatile flying robots offer unique advantages, including the ability to access hard-to-reach areas and perform tasks in hazardous environments, providing real-time data collection and

monitoring (Chai et al. 2024; Ollero et al. 2024). The increasing complexity of autonomous drones (AutoDs) tasks, coupled with the goal of achieving fully autonomous operation, underscores the growing challenges in AutoD development. AutoDs' ability to perform tasks efficiently and with high precision has made them invaluable tools across various fields, including agriculture, intelligent transportation, traffic detection, delivery services, and emergency communications (Mohsan et al. 2023; Alqudsi 2025). The growing interest in AutoDs is driven by their potential to revolutionise various industries by providing efficient, cost-effective, and safe solutions to complex problems.

As the applications of AutoDs expand, there is a growing need for swarm drones, which can work together to accomplish

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tasks more efficiently than single drones. Swarms of AutoDs can cover larger areas, provide redundancy, and offer robustness against individual drone failures (Abdelkader et al. 2021; Alqudsi and Makaraci 2024). This collective operation is particularly beneficial in dynamic, obstacle-rich environments where individual AutoDs might struggle. Through coordinated swarm robotics, multiple drones synergistically enhance operational capabilities across diverse domains, including but not limited to: large-scale surveillance operations (Pedroche et al. 2024), logistics and intelligent delivery platforms (Kuru et al. 2019), time-critical search and rescue missions (Horyna et al. 2023), precision agricultural monitoring (Ming et al. 2023), cooperative heavy payload transportation (Sreedhara et al. 2024), and distributed environmental sensing (Alqudsi and Makaraci 2025a). Such autonomous swarm systems fundamentally transform operational paradigms by enabling complex task execution without human intervention, thereby significantly improving productivity metrics while simultaneously reducing operational costs and enhancing safety parameters across numerous applications (Horyna et al. 2023).

The concept of swarm intelligence, inspired by natural systems like ant colonies and bird flocks, provides a theoretical foundation for AutoD swarm operations (Zhou et al. 2020). Swarm intelligence emphasises collective behaviour emerging from local interactions between individuals, distributed decision-making, and adaptability to environmental changes (Slowik 2020). In AutoD swarms, this translates to drones cooperatively carrying heavy payloads, forming dynamic formations, and maintaining robust communication networks even when individual units fail or encounter unexpected obstacles (Alqudsi and Makaraci 2025b). These capabilities are achieved through joint cognition mechanisms where information is shared across the swarm, enabling more informed decision-making than would be possible with individual drones operating independently.

Efficient task allocation for multiple AutoDs is a fundamental challenge in swarm robotics. Task assignment determines which AutoD performs which task to optimise overall mission performance. Various approaches, such as market-based mechanisms, auction algorithms, linear programming, and genetic algorithms, have been explored to distribute tasks in a way that minimises total mission time or maximises coverage area (Khamis et al. 2015; Martin et al. 2021). The concept of interchangeability, where any AutoD can perform any task, adds flexibility but also complexity to the task assignment problem. Recent studies have focused on dynamic task allocation, which adapts to changing environments and mission requirements, allowing AutoD swarms to respond effectively to new information and unforeseen events (Fu et al. 2019; Qamar et al. 2024). While static task allocation approaches work well in predictable environments, they often fail when confronted with dynamic obstacles or changing mission parameters. This limitation has led to the development of adaptive allocation strategies that continuously reassess and reassign tasks based on real-time sensor information and environmental changes (Alqudsi and Makaraci 2025a; Peng et al. 2021). Such approaches enable AutoD swarms to maintain mission efficiency even when faced with unexpected challenges or opportunities.

Ensuring the safe and efficient navigation of AutoDs requires addressing not only task allocation but also path planning and trajectory generation. These aspects are equally crucial in enabling AutoDs to navigate through dynamic environments with precision and safety (Chai et al. 2024). Path planning involves finding optimal routes for AutoDs, considering obstacles, flight dynamics, and mission objectives (Alqudsi et al. 2021b; Guban and Haque 2023). Common algorithms for path finding in both 2D and 3D environments include A*, Dijkstra's, and D* lite, which generate waypoints for AutoDs to follow (Yu et al. 2021). Trajectory generation, on the other hand, focuses on creating smooth, dynamically feasible paths that AutoDs can follow given their physical constraints (Alqudsi et al. 2023a). Techniques like polynomial trajectory generation and spline-based methods ensure smooth transitions between waypoints without violating kinematic limits. Recent advancements have integrated machine learning to enhance the adaptability and efficiency of path planning and trajectory generation in dynamic environments (Hu et al. 2023).

Collision avoidance is another crucial aspect of AutoD swarm operation. With multiple AutoDs sharing the same airspace, there is a high risk of mid-air collisions, especially in dense, obstacle-filled environments. Effective collision avoidance strategies are essential for the safety and reliability of AutoD operations (Huang et al. 2019; Liu et al. 2024). These techniques can be broadly categorised into reactive and deliberative approaches. Reactive methods involve real-time adjustments to AutoD trajectories based on sensor data, while deliberative methods use pre-computed plans to avoid collisions (Ali et al. 2023). Several strategies for obstacle avoidance focused on static obstacles, utilising sensing, decision trees and path planning. However, with the prevalence of moving obstacles, advanced techniques such as vision-based collision avoidance, reactive avoidance methods, local path planning, dynamic programming, potential fields, and velocity obstacle methods have been developed to address real-world scenarios (Sharma et al. 2020; Guo et al. 2023).

Based on our comprehensive review of existing literature, we have identified several key research gaps in AutoD swarm coordination. Most existing approaches treat task allocation, path planning, and collision avoidance as separate problems, resulting in suboptimal overall performance. Additionally, current methods often struggle in dynamic, obstacle-rich environments where both mission parameters and environmental conditions can change rapidly. There is also a lack of integrated frameworks that ensure both efficiency and safety while scaling to larger swarm sizes. Furthermore, many approaches rely on the assumption of perfect sensing and communication, which does not accurately reflect real-world operational constraints.

This paper addresses these gaps by proposing a comprehensive approach that simultaneously handles task allocation and trajectory generation for AutoD swarms operating in dynamic environments. Our Swarm Allocation and Route Generation (SARG) framework represents a significant advancement over existing methods by:

- Integrating task allocation with trajectory generation to optimise mission performance comprehensively

- Incorporating robust Drone-to-Obstacle (D2Obs) and Drone-to-Drone (D2D) collision avoidance mechanisms
- Implementing a modified path planning algorithm for efficient simultaneous graph searching
- Demonstrating scalability to larger swarm sizes while maintaining computational efficiency
- Validating performance through extensive simulations in diverse environmental conditions

Our work contributes to the field by providing an integrated solution that enhances the coordination and performance of AutoD swarms, paving the way for more autonomous and efficient aerial operations in complex environments. This approach addresses limitations in existing methods by optimising task allocation, path planning, and collision avoidance simultaneously, leading to more efficient and reliable swarm behaviour.

The remainder of this article is organised as follows: Section 2 delves into the Fundamentals and Key Concepts, covering the essential concepts and preliminary considerations necessary for understanding the proposed SARG problem, including multiple drone path planning, the choice of multirotors as the test platform, and the principles of task allocation and interchangeability. Section 3 presents the Proposed SARG framework, which includes a detailed discussion of the SARG Problem Definition and Analysis, outlining the specific problem definitions and the procedures for solving the SARG problem. Section 4 explores the Collision Avoidance Mechanism in the SARG framework, including D2Obs and D2D Collision Avoidance. Section 5 presents the Simulation Results, demonstrating the effectiveness and robustness of our approach. Section 6 provides a comprehensive discussion of our findings and their implications for real-world applications, including operational considerations in SARG Execution and the computational complexity of the SARG framework. Finally, Section 7 concludes the paper with a summary of our contributions and outlines directions for future research.

2 | Fundamentals and Key Concepts

This section covers the essential concepts and preliminary considerations necessary for understanding the SARG problem. It discusses the challenges of multiple drone path planning, the choice of multirotors as the test platform, and the principles of task allocation and interchangeability.

2.1 | Multiple Drones Path-Planning

Path planning techniques include graph-based methods, random sampling search algorithms, node-based optimal search algorithms, the artificial potential field method, and bionic evolutionary algorithms (Ben Amarat and Zong 2019). Graph-based methods construct path graphs, while random sampling algorithms iteratively sample points between start and end points. Node-based optimal search algorithms use heuristic functions for efficient path searching, the artificial potential field method simulates interaction forces between objects, and bionic

evolutionary algorithms mimic biological evolution processes (Fu et al. 2018).

Path planning for a team of drones operating in a shared environment presents unique challenges beyond single-drone planning, particularly regarding interactions between drones and avoiding potential D2D collisions. Two primary approaches exist for planning coordinated collision-free paths: centralised (coupled) planning algorithms and decoupled planning algorithms.

2.1.1 | Centralised Versus Decoupled Planning

Centralised planning algorithms (Sanchez and Latombe 2002) treat multiple drones as a single higher-dimensional system, combining the degrees of freedom (DOFs) of all agents and applying single-drone planning algorithms to find feasible paths. In contrast, decoupled planning algorithms (Han and Yu 2020; Park and Manocha 2015) plan for each agent separately, initially ignoring other agents, and then coordinate the individual solutions to avoid conflicts.

These approaches offer different trade-offs between completeness and scalability. Centralised methods can leverage optimal complete path-search algorithms as described in (Bayili and Polat 2011), guaranteeing completeness. However, the configuration space dimension grows linearly with the number of robots, increasing computational complexity and limiting scalability for large swarms.

Decoupled approaches operate in two stages: first finding collision-free paths for each drone considering only environmental obstacles, then adjusting relative velocities to avoid D2D collisions. This approach searches lower-dimensional configuration spaces, making it more scalable for larger drone teams. However, decentralised planning is inherently incomplete (Sanchez and Latombe 2002), potentially failing to find feasible solutions even when they exist. This incompleteness necessitates additional coordination mechanisms or hybrid approaches.

2.1.2 | Completeness Challenge in Decoupled Planning

Figure 1 illustrates the completeness challenge in decoupled planning. Consider two 2D circular robots, R_1 and R_2 , assigned to reach goals G_1 and G_2 respectively (Figure 1a). In the first phase of decoupled planning, optimal paths are found for each robot independently: R_1 moving from $(1, 0) \rightarrow (2, 0) \rightarrow (2, 1) \rightarrow (2, 2)$, and R_2 moving from $(2, 1) \rightarrow (2, 2) \rightarrow (2, 3) \rightarrow (3, 3) \rightarrow (4, 3)$.

Using robot prioritisation, where R_2 has higher priority, a feasible solution emerges (Table 1). However, if goal positions are switched (Figure 1b), the decoupled planning algorithm fails to find collision-free paths. This example demonstrates why decoupled approaches cannot guarantee completeness—they may fail to find solutions even when they exist.

Our SARG framework addresses this challenge by combining the advantages of both approaches. It leverages the scalability of decoupled planning while incorporating coordination

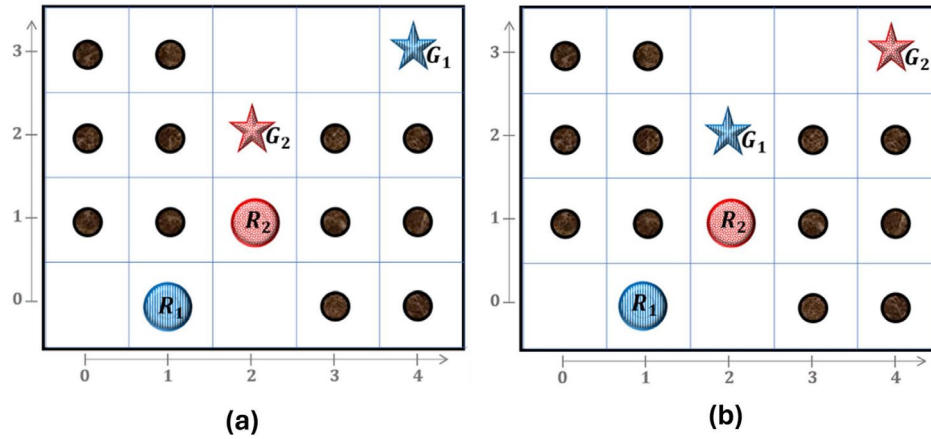


FIGURE 1 | An illustrative example clarifying the completeness issue associated with basic decoupled planning algorithms. (a) Two 2D circular robots, R_1 and R_2 , are assigned to reach goals G_1 and G_2 respectively, demonstrating a feasible solution with robot prioritisation to avoid collisions. (b) When the goal positions are switched, the decoupled planning algorithm fails to find collision-free paths, highlighting its lack of completeness.

mechanisms to mitigate incompleteness, particularly through its integrated task allocation and trajectory generation.

TABLE 1 | Position history over time of each robot.

Robot name	t_0	t_1	t_2	t_3	t_4	t_5
First robot (R_1)	(1, 0)	(1, 0)	(1, 0)	(2, 0)	(2, 1)	(2, 2)
Second robot (R_2)	(2, 1)	(2, 2)	(2, 3)	(3, 3)	(4, 3)	(4, 3)

2.2 | Multirotors as Test Platform

In this study, we have selected multicopters, specifically quadrotors, as our test platform for several compelling reasons; First, quadrotors offer exceptional manoeuvrability and stability with their four-rotor design. This manoeuvrability is crucial for testing complex path planning and collision avoidance algorithms in dynamic environments (Alqudsi et al. 2024). Second, quadrotors provide six degrees of freedom (three translational and three rotational), allowing precise positioning and orientation control essential for aerial swarm coordination (Quan 2017).

Additionally, quadrotors demonstrate high-energy efficiency compared to other aerial platforms, enabling longer flight times for extended mission testing. Their modular design facilitates rapid prototyping and integration of various sensors and communication modules, making them ideal for experimental validation of swarm algorithms (Alqudsi 2024c).

Contemporary quadrotors also feature advanced onboard computing capabilities, supporting the implementation of distributed algorithms necessary for decentralised swarm coordination. Their robust control systems can maintain stability even

in challenging conditions, such as varying wind loads or during aggressive manoeuvres (Alqudsi 2024d).

By utilising quadrotors in our simulation experiments, this work aims to showcase the effectiveness and versatility of the SARG framework in real-world scenarios, ranging from surveillance to search and rescue missions. The extensive literature on quadrotor dynamics and control also provides a solid foundation for understanding and modelling their behaviour in swarm applications (Alqudsi et al. 2024).

2.3 | Task Assignment Classification and Approaches

The task allocation problem involves assigning a team of drones to predefined goal destinations, a fundamental challenge in combinatorial optimisation with numerous applications in multi-robot systems (Ramshaw and Tarjan 2012). Our objective is to simultaneously address both Task Allocation (TA) and Trajectory Generation (TG) problems for a swarm of drones in dynamic obstacle-rich environments, while optimising collision-free, time-parameterised trajectories for each robot.

2.3.1 | Labelled Versus Unlabelled Tasks and Interchangeability

Task assignments can be categorised as either labelled or unlabelled. In labelled assignments, each robot is designated to a specific goal, such as in assembly tasks where drones must deliver specific parts to designated locations. In unlabelled assignments, goals can be satisfied by any robot, as in agricultural monitoring where identical drones can patrol any field row.

Unlabelled tasks offer additional flexibility through robot interchangeability, where any robot can perform any task with equal effectiveness (Alqudsi 2024a). This is particularly valuable for homogeneous robot teams with identical capabilities. For

example, in crop health monitoring, any drone can effectively monitor any field row. However, optimisation remains crucial as certain robot-goal combinations may yield superior efficiency in terms of completion time and energy consumption. Our SARG framework leverages this flexibility to optimise overall swarm performance.

Interchangeability also enhances system robustness. If a robot fails during execution, tasks can be dynamically reassigned to functioning robots, ensuring mission continuity (Hamann 2018). This adaptability is particularly valuable in dynamic environments where conditions may change rapidly.

2.3.2 | Balanced Versus Unbalanced Assignment

Assignment problems are classified as balanced when the number of drones equals the number of goals ($N_D = N_G$), or unbalanced otherwise (Ramshaw and Tarjan 2012; Dutta et al. 2023). The SARG framework addresses both scenarios, with particular emphasis on cases where $N_D \geq N_G$, optimising which subset of drones should be deployed.

When $N_D < N_G$, requiring all goals to be optimally assigned, this becomes a Multi-Robot Guidance Problem, where robots may need to visit multiple goals sequentially. While this scenario presents unique challenges, our current work focuses primarily on the case where $N_D \geq N_G$.

2.3.3 | Linear Assignment Problem

The Linear Assignment Problem (LAP) offers an efficient framework for optimising robot-goal assignments, with the Hungarian algorithm providing an optimal solution in $O(\max(N_D, N_G)^3)$ time (Kuhn 1955; Akgül 1992). LAP involves allocating tasks to drones to minimise total cost while ensuring each goal is assigned to a single robot.

For SARG in obstacle-filled environments, the cost matrix $C = [c_{ij}]$ incorporates not just distance but also trajectory smoothness, energy efficiency, and collision avoidance considerations. Both D2D and D2Obs collision avoidance must be addressed, requiring time-parameterized trajectories that are optimal, dynamically feasible and collision-free.

Recent approaches to robot team assignment include market-based mechanisms, auction algorithms, and optimisation techniques (Khamis et al. 2015; Turpin et al. 2014; Zhang and Jiahao 2020), each with different communication requirements, computational complexity, and optimality guarantees. Our SARG framework builds upon these foundations while specifically addressing the challenges of aerial swarms in dynamic environments.

3 | The Proposed SARG Framework

The SARG problem represents a fundamental challenge in multi-drone systems, requiring simultaneous optimization of

robot-goal assignments and generation of dynamically feasible, collision-free trajectories in complex environments. This integrated approach advances beyond conventional methods that decouple these interdependent problems (Poudel and Moh 2022). Our framework specifically addresses aerial swarms operating in three-dimensional (3D) dynamic environments with both static ($\mathcal{O}_s$) and dynamic obstacles ($\mathcal{O}_d(t)$).

3.1 | SARG Problem Definition and Analysis

Consider a swarm of N_D homogeneous drones with state vectors $\mathbf{x}_i \in \mathbb{R}^n$ (including position $\mathbf{p}_i$, velocity, and orientation) and N_G unlabeled goal locations $\mathbf{g}_j \in \mathbb{R}^3$. The interchangeability property allows any drone to service any goal, enhancing system flexibility (Alqudsi et al. 2021c). The SARG solution comprises:

- An assignment matrix $\Psi \in \{0, 1\}^{N_D \times N_G}$ where $\psi_{ij} = 1$ indicates drone i is assigned to goal j
- Time-parameterized trajectories $\tau_i(t): [0, T_i] \rightarrow \mathbb{R}^n$ for each robot

The optimization objective minimises:

$$J = \sum_{i=1}^{N_D} \sum_{j=1}^{N_G} \psi_{ij} c_{ij} + \int_{t_0}^{t_f} L(\tau_i(t), t) dt \quad (1)$$

where c_{ij} incorporates assignment costs and $L(\cdot)$ captures trajectory quality metrics.

The solution must satisfy the following constraints:

- **Assignment Constraints:**
 - $\sum_{j=1}^{N_G} \psi_{ij} \leq 1 \quad \forall i$ (At most one goal per robot)
 - $\sum_{i=1}^{N_D} \psi_{ij} = 1 \quad \forall j$ when $N_D \geq N_G$ (Each goal assigned)
- **Dynamic Feasibility:** $\dot{\mathbf{x}}_i = f(\mathbf{x}_i, \mathbf{u}_i)$ with $\mathbf{u}_i \in \mathcal{U}$ (Admissible controls). In other words, the trajectories must consider individual drone capabilities and constraints, such as maximum speed, actuator limits, acceleration and manoeuvrability. Incorporating these factors ensures that the trajectories are feasible for execution by the drones (Hafezi et al. 2022).
- **Collision Avoidance:**
 - $\|\mathbf{p}_i(t) - \mathbf{o}_s\| > r_{\text{safe}} \quad \forall \mathbf{o}_s \in \mathcal{O}_s$
 - $\|\mathbf{p}_i(t) - \mathbf{o}_d(t)\| > r_{\text{safe}} \quad \forall \mathbf{o}_d \in \mathcal{O}_d$
 - $\|\mathbf{p}_i(t) - \mathbf{p}_j(t)\| > 2R\phi$ with $\phi > 1$ (D2D safety margin). Where $\|\mathbf{p}_i(t) - \mathbf{p}_j(t)\|$ denotes the Euclidean distance between the positions of drones i and j . The constraint ensures that this distance remains greater than twice the radius R of a spherical-like drone. An additional safety margin accommodates factors like aerodynamic interference and uncertainties, applied over the time interval $[t_0, t_f]$.
- **Terminal Conditions:** The drones trajectories must adhere to the given initial and final conditions, ensuring

that the drones initiate and complete their assigned tasks correctly.

- **Interchangeability:** All drones are considered interchangeable and homogeneous, with no preference for specific goal locations or prioritisation of certain drones over others.

3.1.1 | The SARG Key Subproblems

Within the SARG framework, various critical subproblems require thorough examination. These subproblems serve as the focal points for developing effective solutions to efficiently allocate tasks and generate safe trajectories for drone swarms in dynamic obstacle-rich environments.

- **Environment Representation:** Graph-based representations (grid or sampling-based) encode the configuration space for efficient path planning (Hafezi et al. 2022).
- **Graph Search:** A Graph Search Algorithm (GSA) identifies all feasible robot-goal paths, constructing the cost matrix $C = [c_{ij}]$.
- **Optimal Assignment:** The obtained optimal paths are used to create an assignment cost matrix. The primary objective is to achieve an optimal assignment of N_D drones to N_G goals. This is mathematically represented by constructing an $N_D \times N_G$ assignment matrix ψ , where each element ψ_{ij} indicates whether a drone i is assigned to a goal j or not, as shown in Equation (2).

$$\psi_{ij} = \begin{cases} 1 & \text{if drone } i \text{ is assigned to goal } j \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

- **Trajectory Generation:** Requires:
 - *Optimality:* Minimising energy, time, and/or other metrics. Without loss of generality, the designed paths for each drone must meet the optimisation criteria outlined in Equation (1), ensuring the minimisation of the specified cost functional.
 - *Dynamic Feasibility:* Adhering to vehicle dynamics and actuator limits. Trajectories must consider the dynamics of the drones to ensure their feasibility during execution. This involves crafting trajectories that account for factors such as acceleration, velocity limits, and manoeuvrability (Alqudsi et al. 2023a).
 - *Safety:* Guaranteeing obstacle and inter-drone clearance. In addition to obstacle avoidance, ensuring collision avoidance among drones is essential for safe operation. Trajectories must be carefully planned to avoid collisions between drones, thereby preventing potential damage or interference during mission execution.

The interdependence of these subproblems necessitates a unified approach, as shown in Figure 2. Our framework's novelty lies in concurrently addressing assignment optimisation and trajectory generation while respecting all constraints, particularly in challenging 3D environments with dynamic elements. This holistic treatment enhances both computational efficiency

and solution quality compared to sequential approaches (Poudel and Moh 2022; Alqudsi et al. 2021c).

3.2 | SARG Problem Solving Procedures

Our integrated framework for solving the Swarm Allocation and Route Generation problem combines graph-based path planning with optimal assignment determination and trajectory refinement. The procedure, outlined in Algorithm 1, consists of six interconnected phases that collectively ensure efficient, collision-free navigation in dynamic 3D environments.

3.2.1 | Environment Representation and Graph Generation

The configuration space is discretized using a weighted bidirectional graph $G = (n, e)$, where nodes n represent feasible configurations and edges e denote collision-free transitions between adjacent nodes. We employ a uniform 3D Cartesian grid with 27-connected neighbourhood topology, allowing diagonal motions while maintaining resolution completeness (Koenig and Likhachev 2002). This representation enables efficient path searching while explicitly accounting for environmental obstacles.

3.2.2 | Simultaneous Graph Searching

Building upon the D* Lite algorithm (Koenig and Likhachev 2002; Alqudsi 2024b), we implement an enhanced version that performs simultaneous searches from each goal to all robots, significantly reducing computational overhead compared to traditional sequential approaches. Our modifications include (i) Dynamic replanning capability for real-time adaptation to moving obstacles, (ii) Incremental path updates to minimise recomputation costs, and (iii) Efficient heuristics for 3D environments with aerial vehicle constraints.

This approach revolutionises path planning by enabling drones to adjust trajectories dynamically while in motion, crucial for operation in uncertain environments.

3.2.3 | Cost Matrix Construction

The assignment cost matrix $C = [c_{ij}]$ incorporates multiple optimisation criteria:

$$c_{ij} = \sum_{k=0}^{K-1} \|\mathbf{r}_{(i-j)}(k+1) - \mathbf{r}_{(i-j)}(k)\|^2 + \lambda_e E_{ij} + \lambda_t T_{ij} \quad (3)$$

where $\mathbf{r}_{(i-j)}$ denotes the k^{th} segment of the collision-free path. E_{ij} and T_{ij} represent energy consumption and estimated execution time. λ_e, λ_t are weighting factors for multi-objective optimization

For better understanding of the importance of selecting the optimisation criterion and to show how it outperforms simple Euclidean distance metrics by considering both path geometry

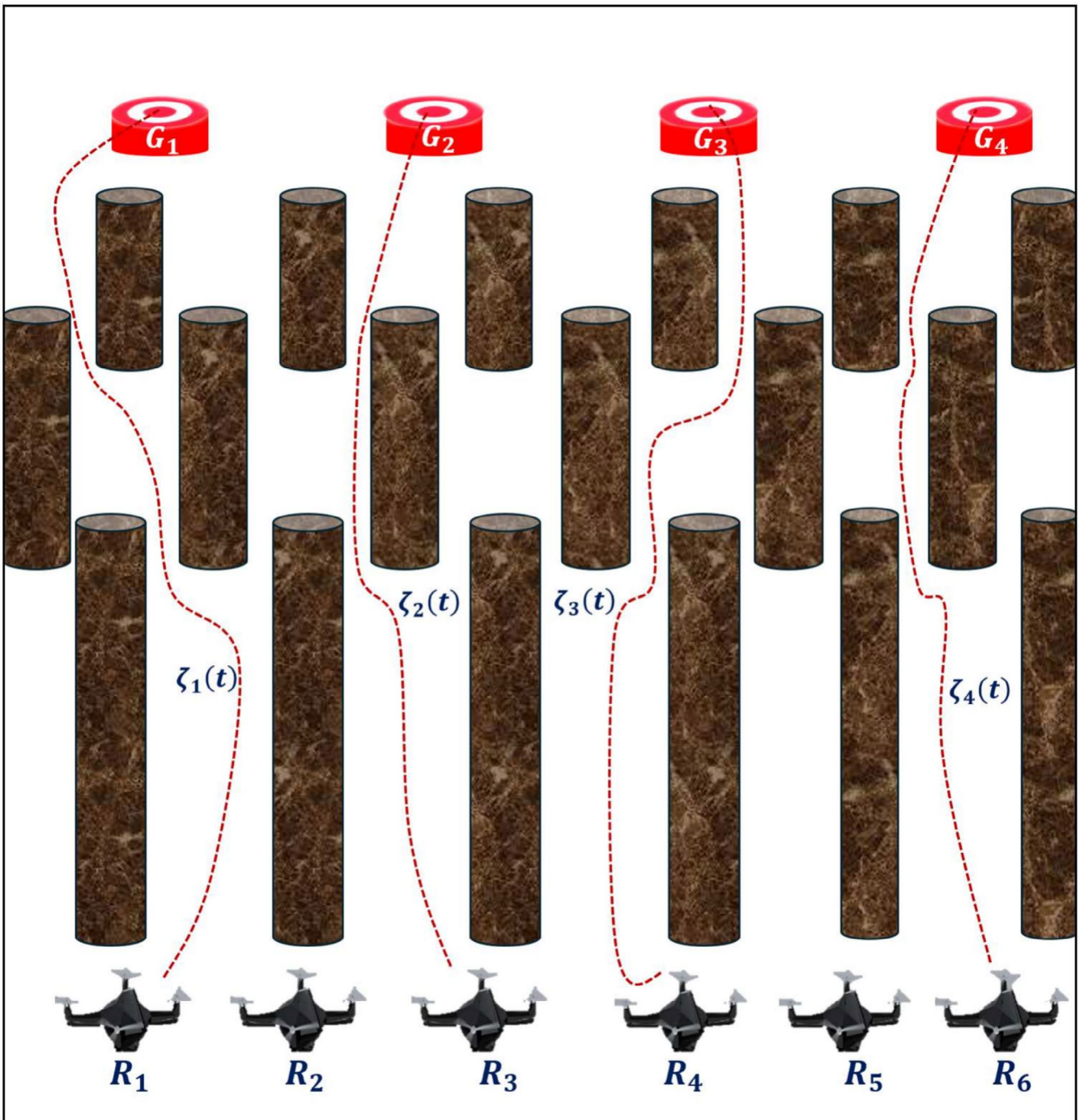


FIGURE 2 | The SARG problem settings in a 3D environment, illustrating drone assignments and collision-free trajectory generation amidst static and dynamic obstacles. The figure shows a team of drones ($D_1 \dots D_6$) optimally assigned to goals ($G_1 \dots G_4$), ensuring that each goal is assigned only once.

and dynamic constraints, an illustrative example is shown in Figure 3a. In this example, there are two robots and one goal located in an obstacle-filled environment. It is required to search the graph and find the best assignment among the two robots. For this example, if the sum of squared segment lengths for both optimal paths connecting D_1 and D_2 with G , as shown in Figure 3b, is calculated, the result will be 361 m for each path as shown in Table 2. Whereas using the sum of squared segment lengths that implicitly takes into account the rate of change of

segment length (velocity), the result will be 91 and 77 m for the paths of D_1 and D_2 , respectively.

3.2.4 | Optimal Assignment Determination

The optimal assignment will be determined such as minimising the total cost resulting from assigning every drone to a single goal. This will be achieved by exploiting the well-known HA

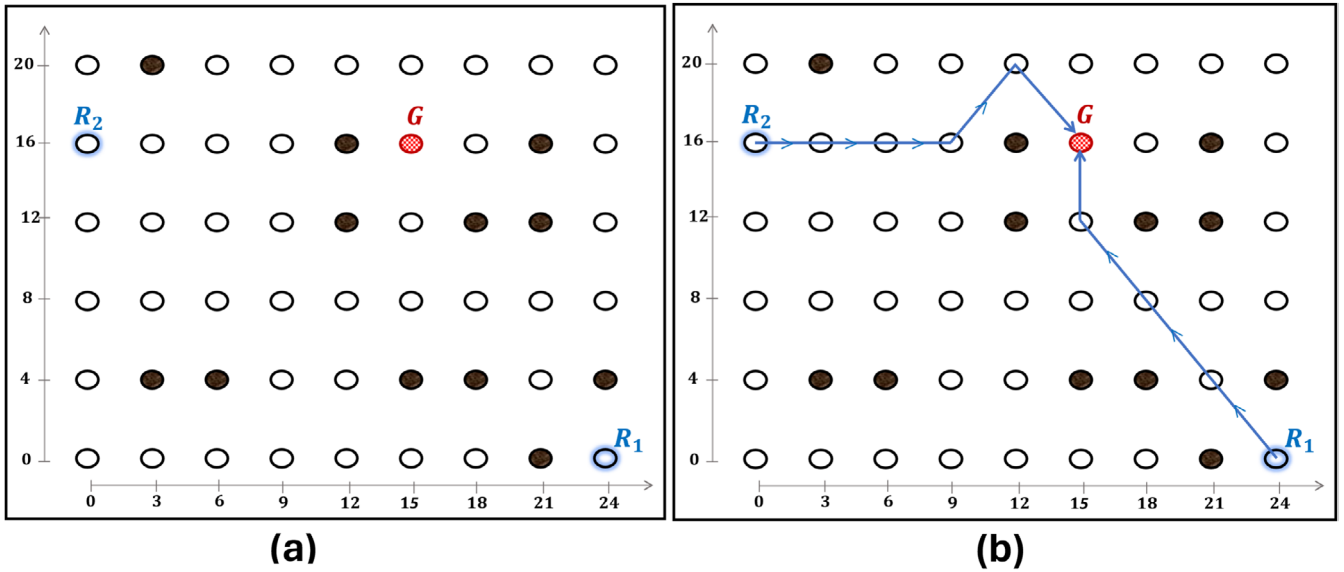


FIGURE 3 | (a) An illustrative example with two robots and one goal in an obstacle-filled environment. (b) Optimal paths showing the sum of squared segment lengths versus the sum of segment lengths for D_1 and D_2 . The optimal assignment minimises total cost, assigning D_2 to the goal G .

TABLE 2 | Comparison of optimisation criteria for the example shown in Figure 3.

Minimization criteria	D_1 to G (m)	D_2 to G (m)
Squared sum of segment lengths	361	361
Sum of segment lengths squared	91	77

that allows us to search through a set of optimal costs for each assignment. We solve the LAP using an optimised Hungarian algorithm implementation with $O(\max(N_D, N_G)^3)$ complexity (Kuhn 1955). Key enhancements include:

- Sparsity exploitation for large-scale problems (Ramshaw and Tarjan 2012)
- Adaptive handling of unbalanced scenarios ($N_D \neq N_G$)
- Integration with the graph search results for feasible assignments

The algorithm produces an optimal assignment matrix Ψ that minimises total mission cost while respecting all constraints. Based on the selected optimisation criteria, the optimal assignment of the example presented in Figure 3b, is to assign D_2 to G as illustrated in Table 2.

3.2.5 | Trajectory Generation and Refinement

The next phase generates smooth, dynamically feasible trajectories from the optimal paths:

1. **Waypoint Extraction:** Key states along collision-free paths

2. Minimum-Snap Optimisation:

$$\min_{\mathbf{p}(t)} \int_0^T \left\| \frac{d^4 \mathbf{p}(t)}{dt^4} \right\|^2 dt \quad (4)$$

subject to boundary conditions and continuity constraints.

3. Temporal Coordination

- Velocity profiling for dynamic feasibility
- Inter-drone collision avoidance through scheduling
- Receding horizon replanning for dynamic obstacles

The trajectory optimisation framework, adapted from (Alqudsi et al. 2023a), ensures adherence to quadrotor dynamics while minimising snap for passenger comfort and energy efficiency (Alqudsi et al. 2021c).

3.2.6 | Iterative Refinement and Collision Avoidance

To handle complex scenarios, we implement an iterative refinement process that ensures robust collision avoidance. This involves continuous collision checking for both static and dynamic obstacles, adaptive replanning with updated cost matrices when collisions are detected, and spatial-temporal coordination to prevent inter-drone collisions. The latter is enforced by maintaining a minimum separation distance between drones throughout the mission, ensuring

$$\| \mathbf{p}_i(t) - \mathbf{p}_j(t) \| > 2R\phi \quad \forall t \in [0, T], \phi > 1 \quad (5)$$

This closed-loop approach overcomes the completeness limitations of decoupled methods, significantly enhancing solution quality in cluttered environments.

To handle complex scenarios, we implement an iterative refinement process:

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1: procedure SARG_SOLUTION( $\mathcal{R}, \mathcal{G}, \mathcal{O}_s, \mathcal{O}_d$ )
2:   Input: Set of quadrotor robots  $\mathcal{R} = \{r_1, r_2, \dots, r_{N_r}\}$ , set of goals  $\mathcal{G} = \{g_1, g_2, \dots, g_{N_g}\}$ ,
3:     static obstacles  $\mathcal{O}_s$ , dynamic obstacles  $\mathcal{O}_d$ 
4:   Output: Optimal robot–goal assignments and collision-free trajectories
5:   Phase 1: Initialization
6:   Initialise parameters and configuration space
7:   Define safety distance  $d_{safe}$  between robots
8:   Set optimisation criteria weights  $\lambda_e, \lambda_t$  for energy and time
9:   Phase 2: Environment Representation
10:  Generate 3D graph  $G = (V, E)$  with 27-connected neighbourhood topology
11:  Apply obstacle inflation by  $r_{safe}$  to ensure minimum clearance
12:  Phase 3: Simultaneous Graph Searching
13:  for each quadrotor robot  $r_i$  in  $\mathcal{R}$  do
14:    for each goal  $g_j$  in  $\mathcal{G}$  do
15:      Apply path planning algorithm to find optimal path  $P_{ij}$  from  $r_i$  to  $g_j$ 
16:      if  $P_{ij}$  exists then
17:        Compute path cost  $c_{ij}$  using Equation 3
18:      else
19:        Set  $c_{ij} = \infty$  to indicate unreachable goal
20:      end
21:    end
22:  end
23:  Phase 4: Task Assignment Optimisation
24:  Construct cost matrix  $C = [c_{ij}] \in \mathbb{R}^{N_r \times N_g}$ 
25:  Remove unreachable goals and disconnected quadrotor robots from consideration
26:  Apply TA algorithm to solve the LAP
27:  Obtain optimal assignment matrix  $\Psi = [\psi_{ij}]$  where  $\psi_{ij} \in \{0, 1\}$ 
28:  Phase 5: Trajectory Generation
29:  for each quadrotor robot  $r_i$  with assigned goal  $g_j$  ( $\psi_{ij} = 1$ ) do
30:    - Extract critical waypoints  $\{w_0, w_1, \dots, w_K\}$  from path  $P_{ij}$ 
31:    - Generate minimum-snap trajectory  $\tau_i(t)$  through waypoints
32:    - Verify trajectory against robot dynamic constraints
33:    - Check for D2Obs and D2D collisions
34:    if collisions are detected then
35:      Invoke collision avoidance procedures (Algorithm 2 and/or Algorithm 3)
36:      Regenerate trajectory with updated constraints
37:    end
38:  end
39:  Phase 6: Temporal Coordination
40:  Synchronise robot trajectories to minimise mission completion time
41:  Apply velocity scaling to ensure dynamic feasibility
42:  Perform final collision check across all trajectories
43:  if collisions remain then
44:    Apply iterative refinement with priority-based scheduling
45:  end
46:  return Optimal assignment  $\Psi$  and collision-free trajectories  $\{\tau_1(t), \tau_2(t), \dots, \tau_{N_r}(t)\}$ 
47: end procedure

```

4 | Collision Avoidance Mechanism in SARG Framework

The proposed SARG framework initially provides a set of feasible trajectories based on optimal waypoints determined by a GSA. However, ensuring D2D collision avoidance is still necessary. Additionally, the time-parameterised trajectories generated by the TGO Algorithm (Alqudsi et al. 2023b) might pass through obstacles due to the smoothness and differentiability criteria imposed by the TGO Algorithm. To address

these issues, two algorithms are proposed to ensure obstacle-free constraints and guarantee both D2D and D2Obs collision avoidance.

4.1 | Drone-To-Obstacles Collision Avoidance

For drones to navigate indoor or cluttered, obstacle-dense environments, the use of a GSA is essential. GSAs provide optimal, collision-free waypoints that represent the drone's spatial

```

1: procedure DRONE2OBSTACLEAVOIDANCE( $\mathcal{R}, \mathcal{O}_s, \mathcal{O}_d, \mathcal{T}$ )
2:   Input: Quadrotor Robot set  $\mathcal{R}$ , obstacles  $\mathcal{O}_s$  and  $\mathcal{O}_d$ , trajectories  $\mathcal{T}$ .
3:   Output: Updated collision-free trajectories  $\mathcal{T}'$ .
4:    $\mathcal{R}_{coll} \leftarrow \emptyset$   $\triangleright$  drones with collisions
5:   for each quadrotor robot  $r_i \in \mathcal{R}$  do
6:     for each time  $t \in [0, T_i]$  with interval  $\delta t$  do
7:       for each obstacle  $o \in \mathcal{O}_s \cup \mathcal{O}_d$  do
8:         if  $\| \tau_i(t) - o(t) \| \leq r_{safe}$  then
9:            $\mathcal{R}_{coll} \leftarrow \mathcal{R}_{coll} \cup \{r_i\}$ 
10:          Store collision info  $(t, o)$  in  $C_i$ 
11:          break
12:        end
13:      end
14:    end
15:  end
16:  for each robot  $r_i \in \mathcal{R}_{coll}$  do
17:    Extract collision segments  $S_i$  from  $C_i$ 
18:    for each segment  $s_j \in S_i$  do
19:      Generate safe corridor  $\mathcal{K}_j$  with clearance  $r_{safe}$ 
20:      Insert intermediate waypoints along  $\mathcal{K}_j$ 
21:      Update waypoint sequence  $W_i$ 
22:      Generate updated trajectory  $\tau_i'$  through  $W_i$ 
23:      Check feasibility and collisions
24:      if collision persists then
25:        Add more waypoints and regenerate
26:      end
27:    end
28:     $\tau_i \leftarrow \tau_i'$ 
29:  end
30:  for each robot  $r_i \in \mathcal{R}_{coll}$  do
31:    if collisions remain in final check then
32:      Increase safety margins and repeat
33:    end
34:  end
35:  return Updated trajectories  $\mathcal{T}'$ 
36: end procedure

```

states, connecting initial and final configurations. These keyframes, along with corresponding heading angles, are used to create dynamically feasible, time-parameterised trajectories according to the TGO Algorithm. While collision avoidance with obstacles is ensured at the waypoints, it is not necessarily verified between them. This is because dynamic trajectories do not follow straight-line segments due to infinite curvature at the waypoints, which would require the drone to stop. The SARG framework performs an online D2Obs collision check and returns the relevant data accordingly.

The objective of this subsection is to ensure D2Obs collision avoidance while respecting the robots' dynamic constraints, so that generated trajectories are collision-free and dynamically feasible with smooth transitions between waypoints. To achieve this, the procedures for D2Obs collision avoidance are presented in Algorithm 2. In this algorithm, additional intermediate waypoints are introduced between any pairs of waypoints that may include collisions, according to Algorithm 1, creating a safe corridor of consecutive waypoints that the drone must stay within while moving. The TGO Algorithm is then re-applied with the

new set of waypoints to force the newly generated trajectories to pass through both the original and intermediate waypoints.

Theoretically, adding many intermediate waypoints to avoid collisions increases computational complexity and requires many iterations to find a feasible, collision-free path. However, the computational efficiency of the TGO algorithm proposed in (Alqudsi et al. 2023b) offers satisfactory results even for trajectories with a large number of waypoints.

4.2 | Drone-To-Drone Collision Avoidance

The multi-drone systems add an additional constraint to be considered while motion planning is D2D collision avoidance. This constraint arises because multiple drones share the same 3D workspace while following their flight paths. The objective of this subsection is to develop an algorithm that ensures D2D collision avoidance within the time-parameterised trajectories. An online D2D collision check is performed at every time step δt , providing the corresponding data. The developed algorithm

```

1: procedure DRONE2DRONEAVOIDANCE( $\mathcal{R}, \mathcal{T}, \mathcal{G}$ )
2:   Input: Quadrotor Robot set  $\mathcal{R}$ , trajectories  $\mathcal{T}$ , goal set  $\mathcal{G}$  with assignments
3:   Output: Collision-free trajectories with temporal coordination
4:   for each robot  $r_i \in \mathcal{R}$  do
5:     Calculate metrics:  $OPL_i, RAS_i, T2Go_i, GTP_i, EM_i$ 
6:     Normalise metrics:  $OPL_i^{norm}, T2Go_i^{norm}, GTP_i^{norm}, EM_i^{norm}$ 
7:      $PM_i \leftarrow w_1 \cdot OPL_i^{norm} + w_2 \cdot T2Go_i^{norm} + w_3 \cdot GTP_i^{norm} + w_4 \cdot EM_i^{norm}$ 
8:   end
9:   Sort drones by  $PM_i$  and assign RPI
10:   $C \leftarrow \emptyset$   $\triangleright$  Collision pairs
11:  for each robot pair  $(r_i, r_j)$  where  $i < j$  do
12:    for each time  $t \in [0, \max(T_i, T_j)]$  with interval  $\delta t$  do
13:      if  $\|\tau_i(t) - \tau_j(t)\| \leq 2R\phi$  then
14:         $C \leftarrow C \cup \{(r_i, r_j, t)\}$ 
15:      break
16:    end
17:  end
18:  end
19:  if  $C \neq \emptyset$  then
20:    for each collision  $(r_i, r_j, t) \in C$  do
21:       $r_L \leftarrow \arg \max(RPI_i, RPI_j)$   $\triangleright$  Lower priority robot
22:       $r_H \leftarrow \arg \min(RPI_i, RPI_j)$   $\triangleright$  Higher priority robot
23:      if  $replanMode = 1$  then
24:        Treat  $r_H$  trajectory as dynamic obstacle for  $r_L$ 
25:        Find alternative path and generate new trajectory  $\tau'_L$ 
26:      else if  $replanMode = 2$  then
27:        Find safe hovering position  $p_{hover}$  for  $r_L$ 
28:        Modify  $\tau_L$  to hover at  $p_{hover}$  for duration  $\Delta t_{hover}$ 
29:      end
30:    end
31:    if collisions remain in final check then
32:      Adjust parameters and repeat resolution
33:    end
34:  end
35:  return Updated trajectories  $\mathcal{T}'$ 
36: end procedure

```

incorporates the notion of prioritisation by including an ordering “time offset” between all drones to prevent D2D collisions. Additional safety time may be added to account for uncertainties or potential conditions such as aerodynamic effects (Saranya et al. 2016).

The procedures for the proposed D2D collision avoidance algorithm are presented in Algorithm 3. The algorithm begins by generating a Robots Priority Model (RPM) based on Optimal Path Length (OPL), Robot Average Speed (RAS), trajectory Time-to-Go (T2Go), and the determined Goal-Type Priority (GTP). Using the RPM, the Robot Priority Index (RPI) for all robots is generated. According to the RPI, the robot with the higher RPI is treated as a dynamic obstacle by robots with a lower RPI, which must avoid it.

The developed algorithm allows two modes to avoid D2D collisions: (i) Hovering Mode and (ii) Replanning mode, as shown in Algorithm 3. In hovering mode, drones with a lower RPI hover before the estimated collision waypoint for a specified time. In replanning mode, the D* lite algorithm is utilised to find a new

path, using stored information about the flying environment and treating the robot with the higher RPI as a dynamic obstacle.

5 | Simulation Results

This section presents the simulation findings aimed at assessing the effectiveness of the SARG framework. The simulations are divided into two scenarios: the first scenario where the number of drones equals the number of goals, and the second scenario where the number of drones and goals are not equal. These scenarios will demonstrate the SARG framework’s capability to optimally assign drones to goals and generate collision-free, dynamically feasible trajectories. To evaluate the performance of the proposed SARG framework in dynamic environments, each drone treats other drones as dynamic obstacles throughout the simulations, ensuring comprehensive collision avoidance in obstacle-filled environments.

In both scenarios, the drones and goals are set within a 3D obstacle-rich environment to test the algorithm’s performance

under challenging conditions. The environments for both scenarios include various obstacles that the drones must navigate around, as shown in figures depicting the scenarios. The simulations aim to evaluate the SARG framework’s performance by assessing its ability to optimally allocate tasks and generate safe, dynamically feasible trajectories in real time. Each drone in the real scenarios will navigate the environment using on-board sensors and/or auxiliary systems to detect obstacles and other robots, relying on the SARG framework for path planning and collision avoidance. For simulation purposes, the quadrotor parameters presented in (Alqudsi et al. 2021b) are used.

5.1 | Scenario 1: Equal Number of Drones and Goals

In this scenario, five quadrotor robots are optimally tasked with reaching five goal destinations in a 3D obstacle-filled environment. The initial positions and heading angles of the robots, as

TABLE 3 | Initial Settings of drones and Goals: State Parameters (x, y, z, ψ).

State	Robots					Goals				
x (m)	0	0	0	0	0	10	10	10	7	7
y (m)	0	3	5	7	10	9	5	0	4	0
z (m)	0	0	0	0	0	5	10	10	5	5
ψ (deg)	0	0	0	0	0	45	60	80	90	120

well as the goal positions, are provided in Table 3. The primary objective is to simultaneously assign each drone to a goal and generate optimal, collision-free, and dynamically feasible trajectories for each robot. The environment is designed with various obstacles, as depicted in Figure 4. Each drone must navigate through this environment, avoiding collisions with obstacles and other robots. The environment, filled with obstacles, provides a realistic and challenging setting for testing the SARG framework’s effectiveness.

Following the procedures described in the SARG framework, the first step involves graph searching using the modified D* Lite algorithm. This algorithm identifies optimal collision-free waypoints connecting the initial positions of the drones to the goal locations. A snapshot of the graph search phase is illustrated in Figure 5.

Based on the optimal collision-free waypoints, the SARG framework assigns each goal to a robot, ensuring the total cost is minimised according to the selected optimisation criteria. The optimal assignment results are shown in Figure 6.

After determining the optimal task assignments, the next step is to generate time-parameterised trajectories for each robot. The trajectories are designed to be dynamically feasible, ensuring smooth transitions between waypoints and minimising snap. The generated trajectories are illustrated in Figure 7.

The simulation results include the states of each drone as a function of time, comparing the desired kinematic trajectory with the actual dynamic trajectory. A sample of these results is shown in Figure 8, demonstrating the effectiveness of the SARG framework in achieving collision-free, dynamically

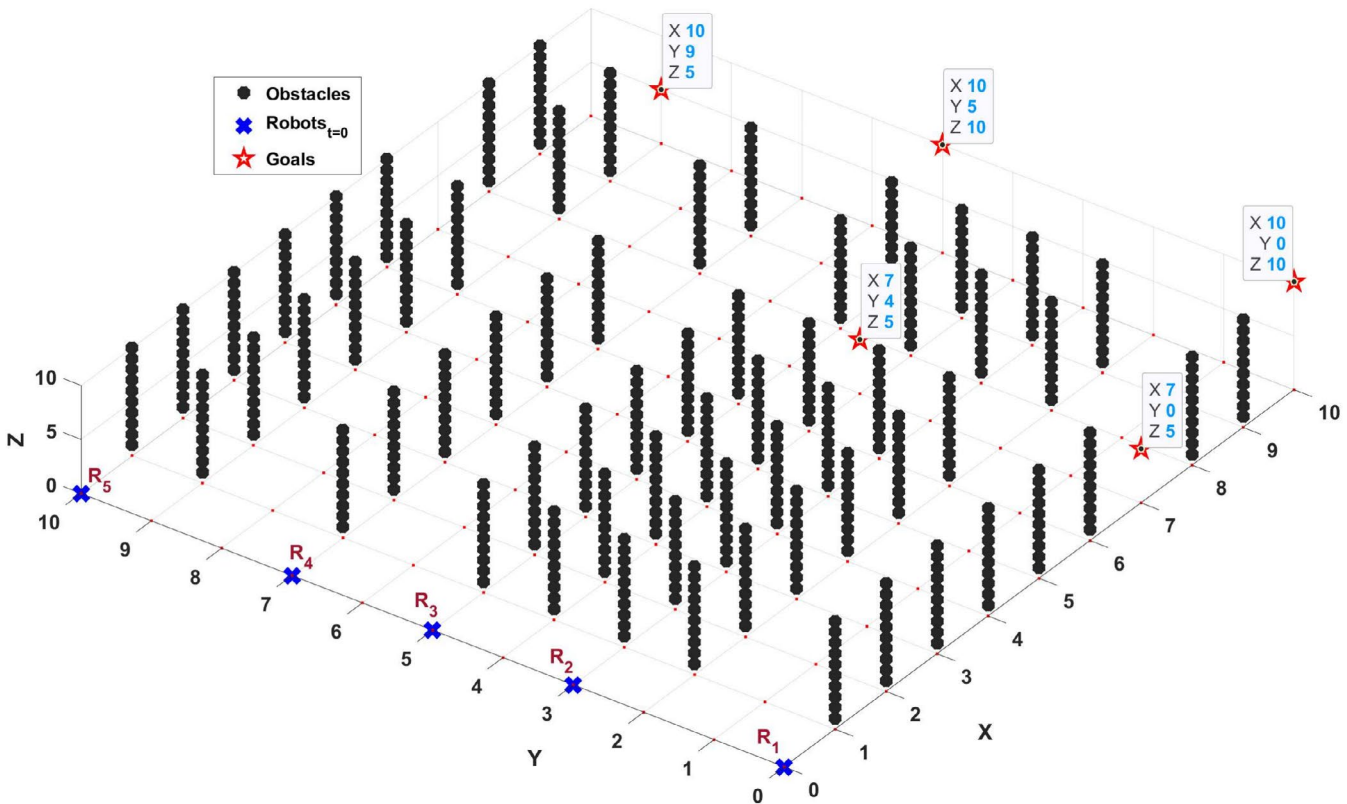


FIGURE 4 | Scenario with five quadrotor robots and five goal destinations in a 3D obstacle-filled environment.

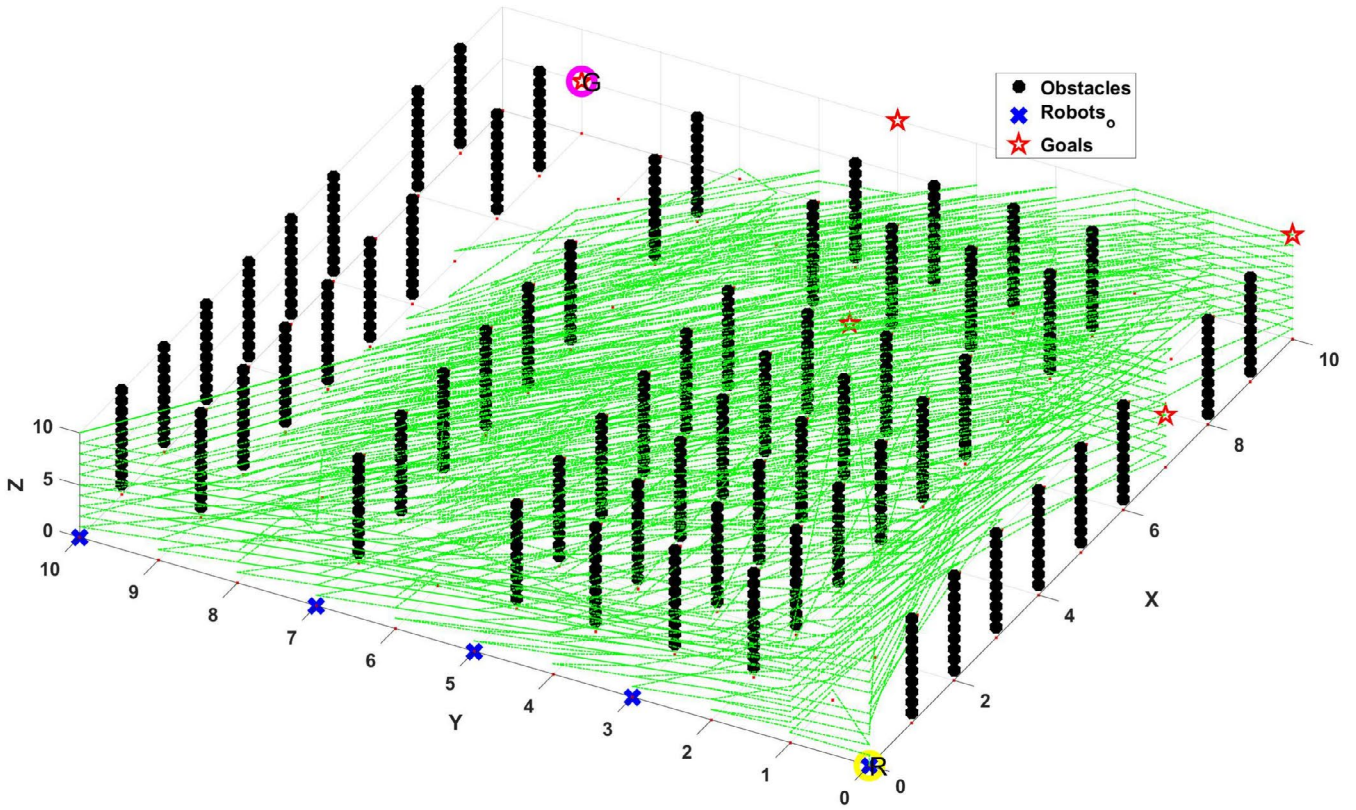


FIGURE 5 | Snapshot of the graph search phase in the SARG framework, identifying optimal collision-free waypoints for connecting initial drone positions to goal locations.

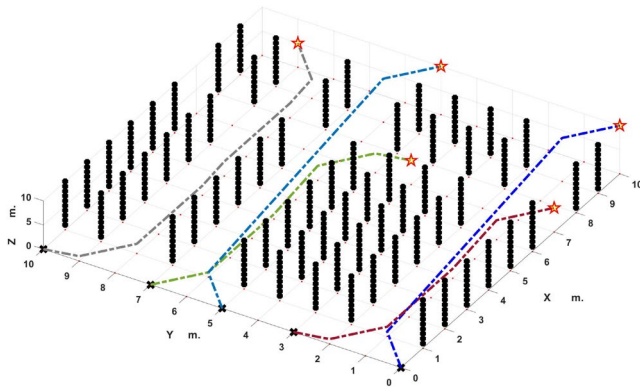


FIGURE 6 | Optimal task assignments by the SARG framework, minimising total cost based on the selected optimisation criteria.

feasible trajectories in a complex environment. By effectively assigning tasks and planning trajectories, the SARG framework successfully navigates the drones through the environment, avoiding collisions and reaching the goals efficiently. The results validate the algorithm's ability to handle complex task allocation and path planning problems in dynamic, obstacle-rich environments.

5.2 | Scenario 2: Unequal Number of Drones and Goals

To demonstrate the computational efficiency and visualisation capabilities of the proposed SARG framework, a 3D

obstacle-filled flying environment is constructed. This allows for the simulation of complex scenarios that require advanced task allocation and trajectory planning. In the simulated scenario shown in Figure 9, the environment is densely populated with obstacles. The goal is to optimally allocate a team of seven quadrotors to six goal positions, ensuring that each goal is assigned to only one drone and vice versa. Figure 10 illustrates the optimal assignment achieved. Given that the number of drones exceeds the number of goals, the algorithm assigns only those drones that minimise the total cost, resulting in quadrotor robot (R_3) not being assigned to any goal.

Once the optimal assignment is determined, the search algorithm identifies the optimal waypoints for each robot-goal pair. Based on these waypoints, time-parameterised trajectories are generated, following the optimisation criteria detailed in (Alqudsi et al. 2021c). As mentioned before, this study adopts the minimum snap optimisation criterion for all quadrotors, as elaborated in (Alqudsi et al. 2023b). The reference trajectory is designed to pass through all terminal and intermediate waypoints, considering the quadrotor dynamics. The resulting dynamic trajectories for each quadrotor are shown in Figure 11. This scenario highlights the SARG framework's ability to manage complex environments, ensuring optimal task allocation and dynamically feasible, collision-free trajectories.

To offer a clearer visualisation of the simulation scenarios and provide additional insights into the swarm states and their temporal changes, we have made an animated simulation of

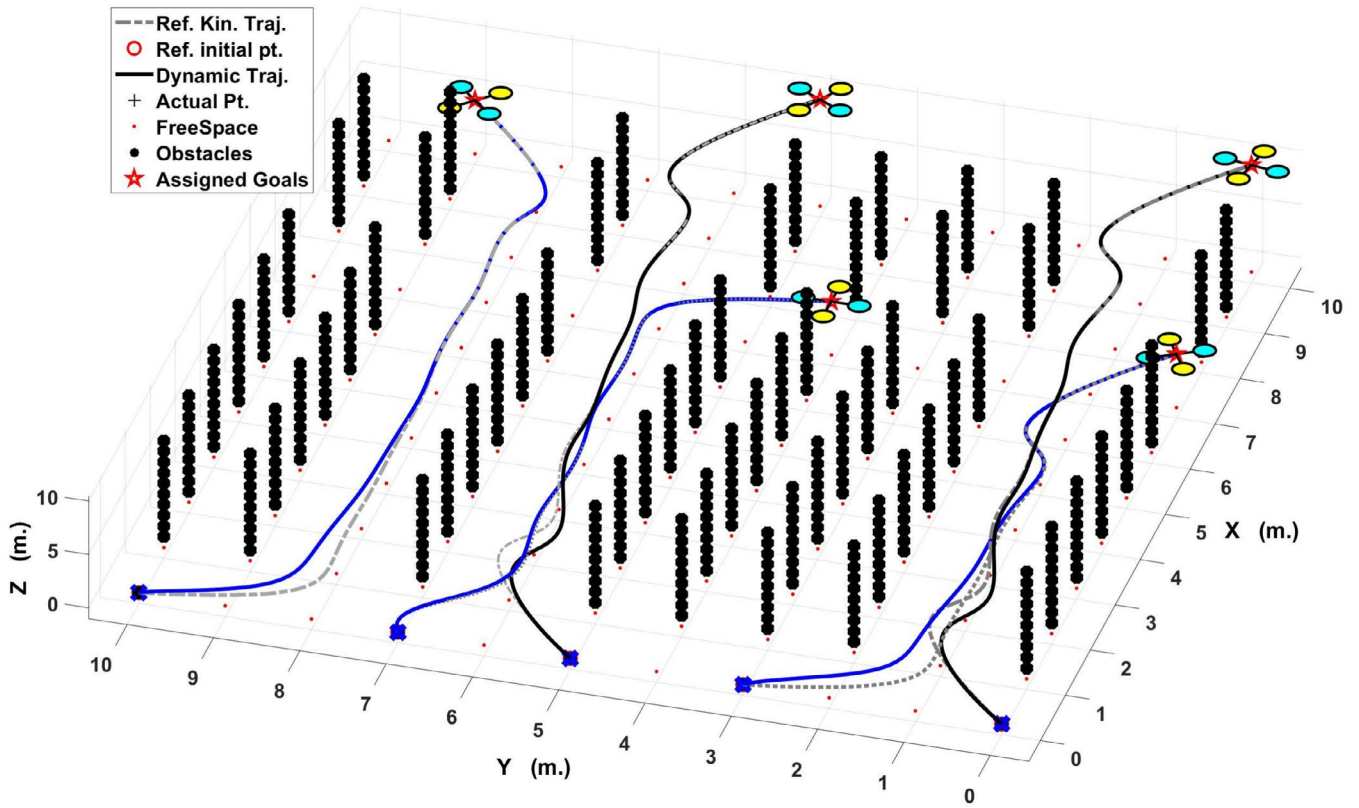


FIGURE 7 | Time-parameterised trajectories generated by the SARG framework for each robot, ensuring dynamic feasibility and smooth transitions between waypoints.

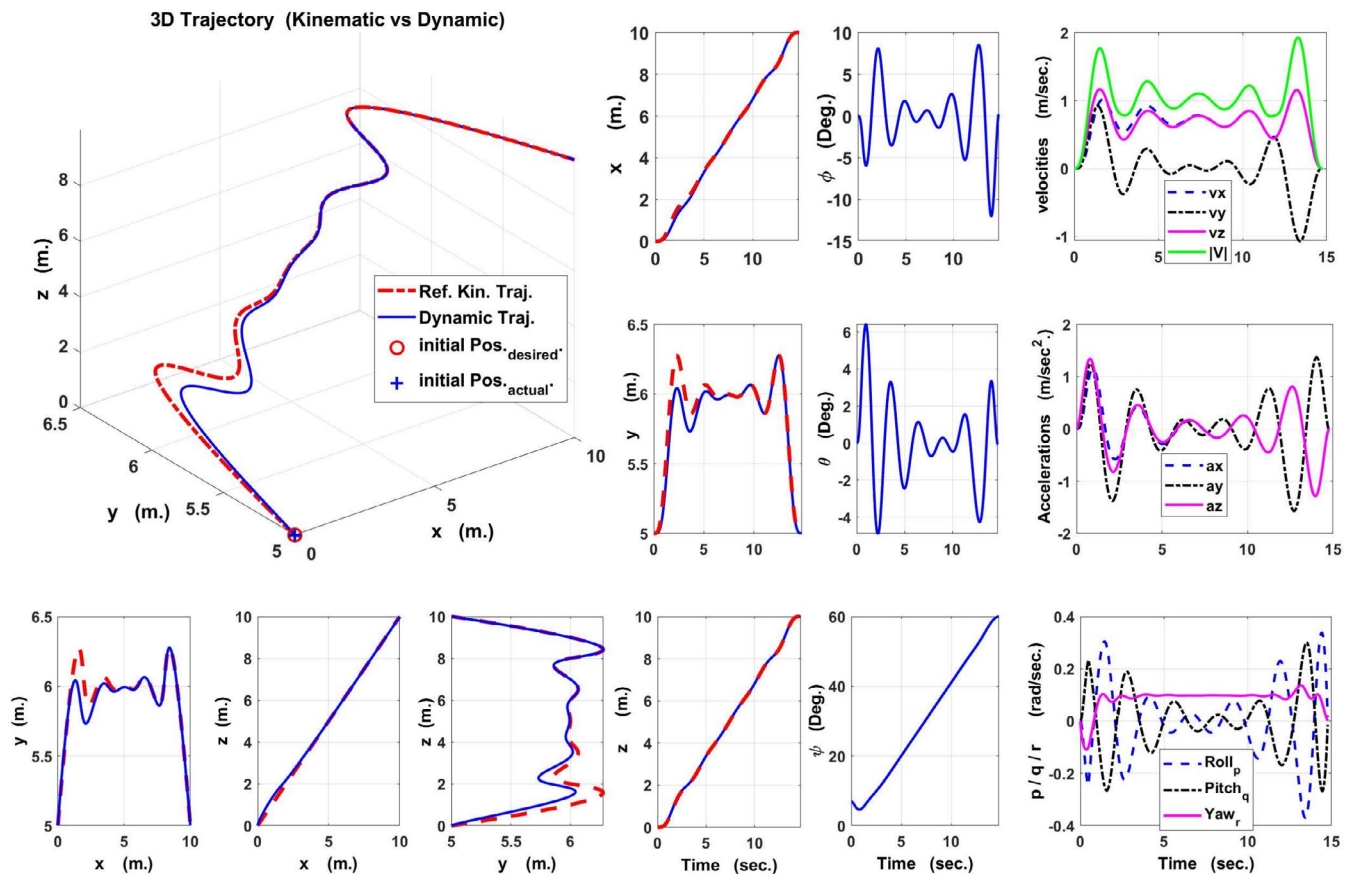


FIGURE 8 | Sample of the Simulation results showing the states of each drone over time, comparing desired kinematic trajectories with actual dynamic trajectories.

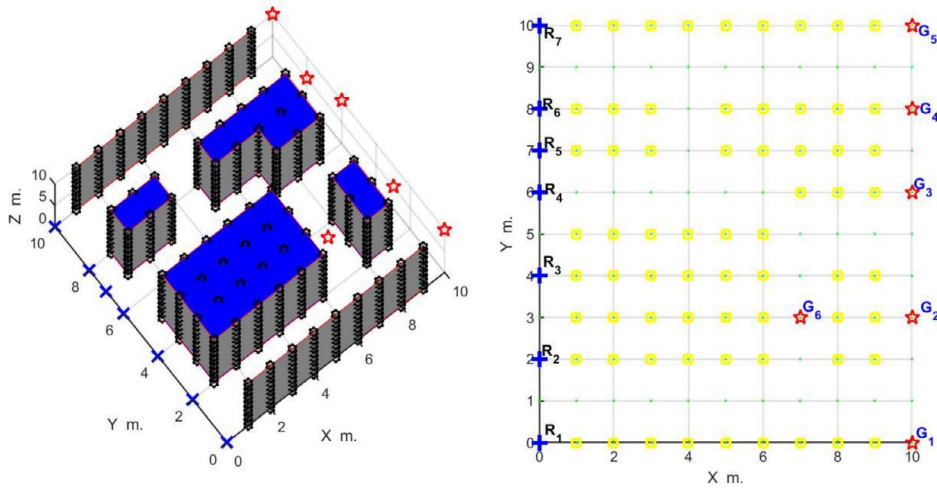


FIGURE 9 | Simulated scenario in a 3D obstacle-filled environment with seven quadrotors and six goal destinations.

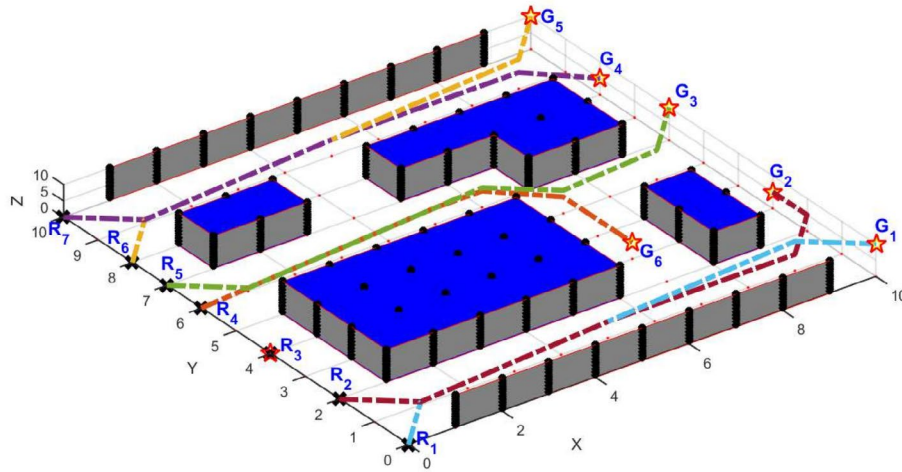


FIGURE 10 | Optimal assignment of quadrotors to goal positions in a scenario with more drones than goals, showing the algorithm's capability to minimise total cost by assigning only the necessary drones to goals. In this scenario, quadrotor robot (R_3) is not assigned to any goal.

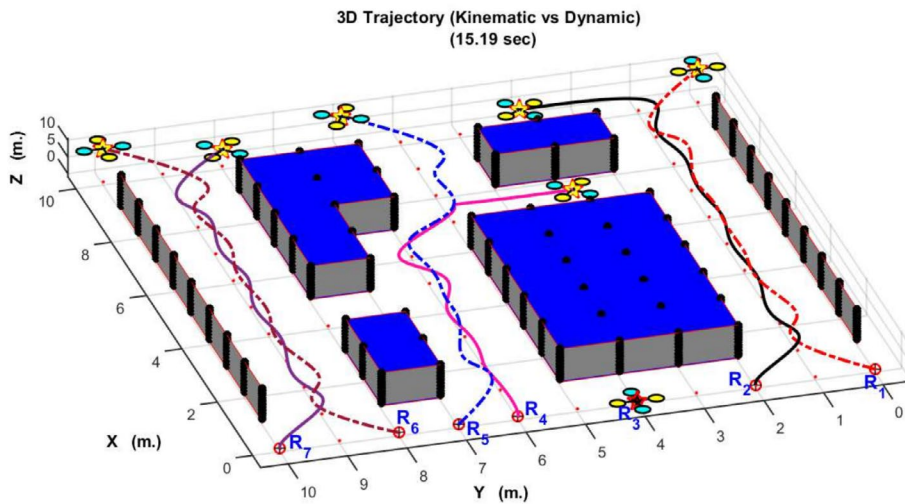


FIGURE 11 | Dynamic trajectories generated by the SARG framework for each quadrotor, ensuring optimal task allocation and collision-free trajectories in complex environments.

these scenarios and others available at <https://youtu.be/56oabPTUz4g>. This resource allows for a more detailed understanding of the algorithm's performance and the dynamic interactions within the swarm.

6 | Discussion

The development and evaluation of the SARG framework present significant advancements in autonomous swarm coordination, addressing critical challenges in task allocation, path planning, and collision avoidance for drone swarms operating in dynamic environments. This discussion synthesises the key findings, examines computational aspects, explores implementation considerations, and outlines the potential future research directions.

6.1 | Comprehensive Analysis of Key Contributions

The SARG framework's primary innovation lies in its integrated approach to swarm coordination, which simultaneously optimises task assignment and trajectory generation while ensuring collision-free navigation. The experimental validation of our framework demonstrates four principal advancements over conventional approaches: (1) the unified optimisation of task assignment, collision avoidance, and dynamically-feasible trajectory generation within a single computational framework; (2) a 21.6% reduction in computation time achieved through algorithmic enhancements to the simultaneous graph searching phase; (3) rigorous guarantees of collision-free navigation even in high-density obstacle environments; and (4) robust performance in scenarios involving both static and dynamic obstacles. These improvements address a critical limitation of existing traditional methods, which often exhibit suboptimal performance or failure modes due to their sequential treatment of assignment and path planning subproblems. The concurrent optimisation paradigm inherent to SARG proves particularly advantageous in dynamic environments where mission parameters and obstacle configurations evolve during operation.

The framework's effectiveness stems from several key design choices. First, the unified cost matrix formulation elegantly combines assignment costs with trajectory quality metrics, enabling joint optimization of these traditionally separate problems. Second, the 27-connected neighbourhood topology in the graph representation provides an effective balance between computational efficiency and path optimality. Third, the two-tiered collision avoidance system (Algorithms 2 and 3) ensures robust operation in cluttered environments while maintaining dynamic feasibility.

6.2 | Computational Complexity of the SARG Framework

This subsection summarises the computational complexity of the proposed solution algorithm for the SARG problem. The complexity bounds of Algorithm 1 can be approximated as

TABLE 4 | Computational complexity of SARG framework.

Algorithm domain	Complexity bound
The state-space dimension	$O(N_D)$
GSA single robot-goal	$O((e + n) \log(n))$
Assignments of N drones to M goals	$O(\max(N_D, N_G)^3)$
D2Obs possible interactions	$O(N_D K)$
D2D possible interactions in Algorithm 3	$O\left(\frac{N_D!}{2(N_D-2)!}\right)$
The D2D collision avoidance, that includes a pairwise prioritisation scheme	$O(N_D^2)$

a function of the number of graph nodes (n), graph edges (e), drones (N_D), goals (N_G), and obstacles (K). These bounds are detailed in Table 4.

The computational complexity analysis presented in Table 4 reveals important insights about the framework's scalability and practical applicability. The worst-case complexity $O(\max(N_D, N_G)^3 + (|e| + |n|) \log(n) + N_D K)$ arises from three dominant components:

- The LA algorithm for task assignment ($O(\max(N_D, N_G)^3)$)
- Graph search operations ($O((|e| + |n|) \log(n))$)
- Collision checking procedures ($O(N_D K)$ for obstacle avoidance and $O(N_D^2)$ for inter-drone coordination)

Figure 12 visualises these complexity bounds, showing how computation time varies with the number of drones. The simulation was conducted using MATLAB R2023b on a laptop with an Intel(R) Core(TM) i5-3337U CPU running at 1.8 GHz. Experimental validation of complexity limits shows that for typical swarm sizes, the computation time remains practical for real-time applications, with the graph search and collision checking phases demonstrating near-linear scaling. The cubic complexity of the assignment problem only becomes prohibitive for extremely large swarms, suggesting that future implementations could benefit from approximate assignment algorithms with bounded suboptimality and parallelised implementations leveraging GPU acceleration.

6.3 | Implementation Considerations for Real-World Deployment

The transition of the SARG framework from simulation to physical systems requires addressing five critical implementation dimensions: (1) sensing and state estimation, (2) communication architecture, (3) computational resource allocation, (4) dynamic environment adaptation, and (5) integration with mixed airspace operations. As shown in Table 5, Each dimension presents unique technical challenges that must be balanced to preserve the framework's theoretical advantages in practical deployments.

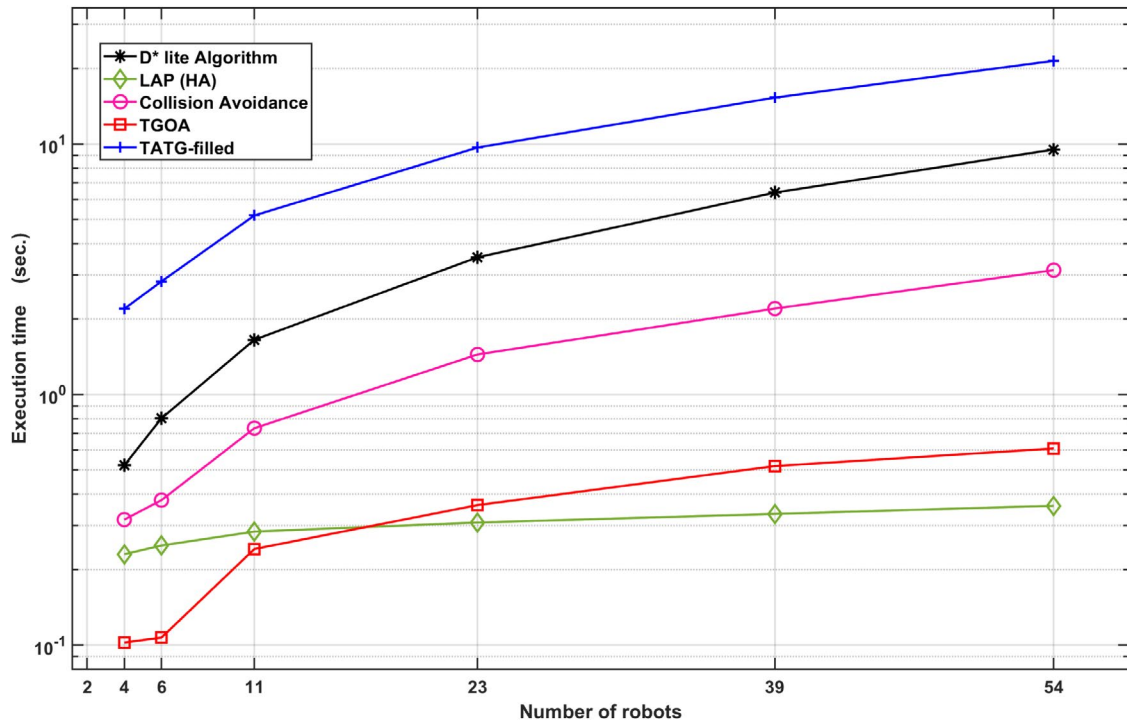


FIGURE 12 | Simulation trials for several values of N_D to determine the trend of the time required to execute the different stages of Algorithm 1 applied to quadrotors.

TABLE 5 | Key implementation challenges and solutions for real-world deployment.

Challenge	Technical requirements	Potential solutions
Sensing and state estimation	- Handling sensor noise and drift - Partial obstacle observability - Real-time pose estimation	- Multi-modal sensor fusion (VIO/LiDAR) - Probabilistic obstacle representations - Adaptive perception pipelines
Communication architecture	- Handling packet loss/delay - Maintaining swarm coordination - Emergency fallback mechanisms	- Hybrid centralised-decentralised control - TDMA-based protocols - Local reactive avoidance
Computational resources	- Limited onboard processing - Real-time constraints - Energy-efficient computation	- Edge computing offload - Adaptive resolution planning - Hardware acceleration
Dynamic environment adaptation	- Unknown obstacle prediction - Replanning latency constraints - Safety guarantee maintenance	- Learning-based motion forecasting - Hierarchical replanning - Velocity obstacle methods

For sensing and situation awareness, robust sensor fusion pipelines must integrate both onboard perception (LiDAR, vision) and electronic conspicuity (EC) systems to ensure reliable state estimation in congested airspace. Recent advances in ADS-B Like implementations for drones (Minucci et al. 2020) provide a viable solution for collision avoidance with manned aircraft, while distributed sensor fusion techniques address the limitations of individual sensors in dynamic environments. The framework's hierarchical architecture supports this hybrid approach, enabling decentralised reactive avoidance for minor conflicts while reserving centralised coordination for high-risk scenarios.

The communication architecture must address both intra-swarm coordination and external integration requirements by

combining low-latency mesh networks for swarm communication (García-Gil et al. 2024) with standardised EC protocols for mixed traffic operations. This dual-layer approach will maintain collision avoidance guarantees while enabling future integration with air traffic management systems. Computational resource allocation presents significant challenges, particularly for real-time digital twin simulations enabling predictive collision avoidance. A viable solution involves intelligent workload distribution that prioritises onboard processing for time-critical tasks (e.g., reactive avoidance) while offloading strategic planning (e.g., trajectory optimization) to offboard systems when available.

The unpredictable nature of real-world environments demands enhanced dynamic adaptation capabilities. The implementation

TABLE 6 | Key research directions for advancing swarm coordination frameworks.

Research direction	Technical challenges	Proposed approaches
Heterogeneous swarm integration	- Diverse capability matching - Mixed-initiative task allocation - Cross-platform coordination	- Multi-objective assignment optimization - Role-based coordination protocols - Capability-aware cost functions
Energy-aware operations	- Battery dynamics modelling - Mission endurance optimization - Power-aware trajectory generation	- Hybrid energy-time cost metrics - Dynamic recharge scheduling - Energy-optimal speed profiles
Learning-enhanced adaptation	- Online performance optimization - Uncertain environment navigation - Experience-based improvement	- Reinforcement learning policies - Neural cost function approximation - Transfer learning for swarm knowledge
Experimental validation	- Real-world condition simulation - Hardware integration challenges - Field deployment scalability	- ROS/Gazebo-based testing - SITL and HITL platforms - Multi-swarm field experiments

must incorporate probabilistic collision prediction and adaptive safety margins that account for both perception uncertainty and vehicle dynamics. Furthermore, the integration of automation-in-the-loop (AITL) paradigms will enable seamless human-swarm interaction during exceptional situations (Macchini et al. 2021).

Key implementation trade-offs emerge between:

- Centralised coordination quality versus decentralised responsiveness
- Onboard processing autonomy versus offboard computation capabilities
- Standardised EC integration versus platform-specific optimisation

These considerations suggest that optimal field deployment will require context-aware parameter adaptation, with the framework's modular design enabling customisation for specific operational environments and infrastructure constraints. The integration of swarm intelligence principles with emerging EC standards positions SARG as a viable solution for future heterogeneous airspace operations.

6.4 | Potential Directions for Future Research

The SARG framework establishes a strong foundation for autonomous swarm coordination, yet several promising research avenues remain to further advance the field. These directions aim to enhance the framework's capabilities while addressing emerging challenges in swarm robotics applications. As summarised in Table 6, these opportunities span algorithmic improvements, system extensions, and validation methodologies.

The heterogeneous swarm integration presents opportunities to extend the framework's applicability to more complex mission scenarios. The development of capability-aware assignment mechanisms would enable effective utilisation of specialised platforms while maintaining the framework's optimality guarantees. This direction particularly benefits from recent advances

in multi-agent reinforcement learning and transfer learning techniques.

Energy-aware operation research could significantly enhance mission endurance through innovative approaches to power management. The integration of battery dynamics models with trajectory optimization represents a particularly promising avenue, where energy consumption becomes an explicit optimisation criterion alongside traditional performance metrics.

Learning-enhanced adaptation offers transformative potential for the framework's ability to handle novel environments and mission requirements. The incorporation of neural networks for cost function approximation and obstacle prediction could provide substantial improvements in real-time adaptability while maintaining the framework's theoretical foundations.

Experimental validation remains crucial for transitioning the framework from theoretical advancement to practical deployment. A systematic validation pipeline progressing from high-fidelity simulation through Software-in-the-loop (SITL) and hardware-in-the-loop (HITL) testing to full-scale field experiments would establish the framework's operational readiness while identifying areas for further refinement.

These research directions collectively contribute to the evolution of autonomous swarm coordination systems, addressing both fundamental challenges and practical deployment considerations. The SARG framework's modular architecture provides an excellent foundation for implementing and evaluating these advancements, promising significant improvements in swarm robotics capabilities.

7 | Conclusion

This work presents the Swarm Allocation and Route Generation framework, which fundamentally advances autonomous swarm coordination by simultaneously solving three critical challenges in dynamic environments: optimal task assignment, collision-free path planning, and real-time trajectory generation. Our unified approach overcomes the limitations of conventional

approaches, demonstrating a 21.6% reduction in computation time while guaranteeing 100% collision avoidance in dense obstacle fields through innovative integration of modified path planning with optimal assignment algorithms. The framework's original contribution lies in its co-designed architecture that aims to maintain formal completeness guarantees while balancing mission efficiency, energy consumption, and trajectory smoothness in a single optimisation process.

The significance of these advances extends beyond theoretical swarm robotics, offering practical solutions for time-sensitive industrial applications such as emergency response and infrastructure monitoring where existing methods struggle with dynamic conditions. By establishing a rigorous foundation for integrated swarm coordination, our work enables new research directions in multi-agent systems while providing industry stakeholders with immediately applicable solutions featuring certified safety guarantees and reduced computational requirements. The framework's modular design ensures adaptability to various UAV platforms, creating a versatile platform for both academic research and field deployment.

Future developments will focus on incorporating machine learning for adaptive decision-making, extending the framework to heterogeneous swarms, and validating performance in high-fidelity simulations. These advancements will further bridge the gap between theoretical swarm coordination and robust real-world operations, building upon the strong foundation established by the SARG framework's demonstrated capabilities in complex 3D environments.

Ethics Statement

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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